

CBSE Class 12 – Mathematics

Linear Programming

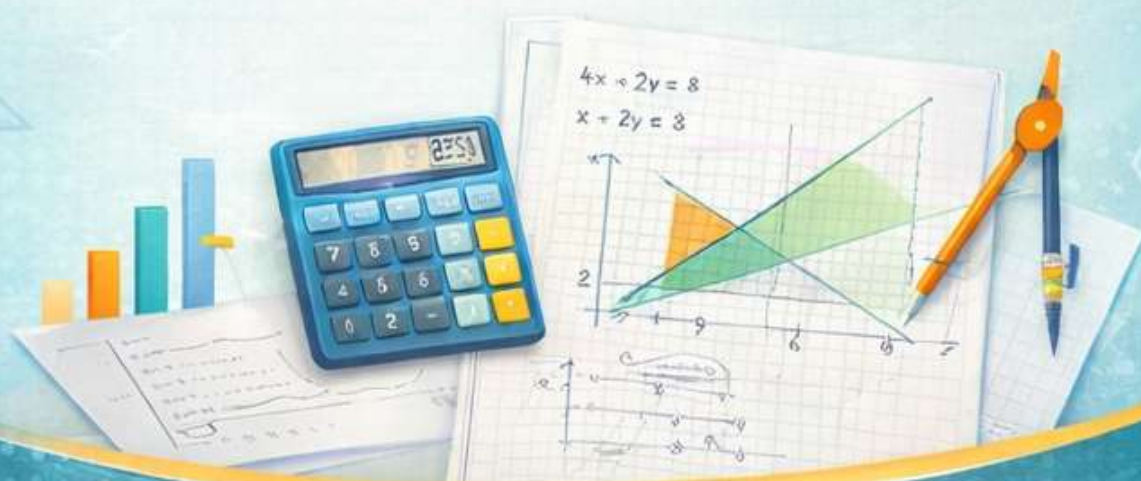


RANJITHAM
STUDY CENTER

CBSE Class 12 | Math Notes

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- ✓ Book Questions Solved



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Linear Programming

Eg: 1 Solve the linear programming problem graphically:

$$\text{Maximize } Z = 4x + y \quad \text{--- (1)}$$

Subject to the constraints

$$x + y \leq 50 \quad \text{--- (2)}$$

$$3x + y \leq 90 \quad \text{--- (3)}$$

$$x \geq 0, y \geq 0 \quad \text{--- (4)}$$

Let us graph the feasible region of the system of inequalities (2), (3) and (4)

from (2) $x + y \leq 50$

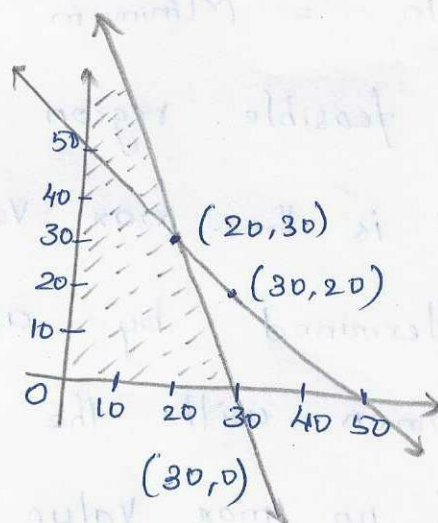
$$x + y = 50$$

$$x = 20$$

$$x = 30$$

$$y = 30$$

$$y = 20$$



from (3) $3x + y = 90$

$$x = 20$$

$$x = 30$$

$$y = 30$$

$$y = 0$$

$$3(0) + 0 \leq 90$$

$$0 \leq 90$$

Now we observe that OABC is feasible region.

Now we use Corner points method,

Corner point Corresponding value of Z

$A(0,0)$ 0

$A(30,0)$ 120

$B(20,30)$ 110

$C(0,50)$ 50

Hence maximum value of Z is 120 at point $(30,0)$

$$Z = ax + by$$

M = Maximum Value

m = Minimum Value

If feasible region is unbounded, then

a) M is the max value of Z , if open half plane determined by $ax + by > M$ has no point in common with the feasible region. Otherwise Z has no max value.

b) m is the min value of Z , if open half plane determined by $ax + by < m$ has no point in common with the feasible region, otherwise Z has no min value.

$$ax + by > M$$

$$ax + by = M$$

Eg: 2

$$Z = -50x + 20y \quad \text{--- (1)}$$

Subject to the Constraints:

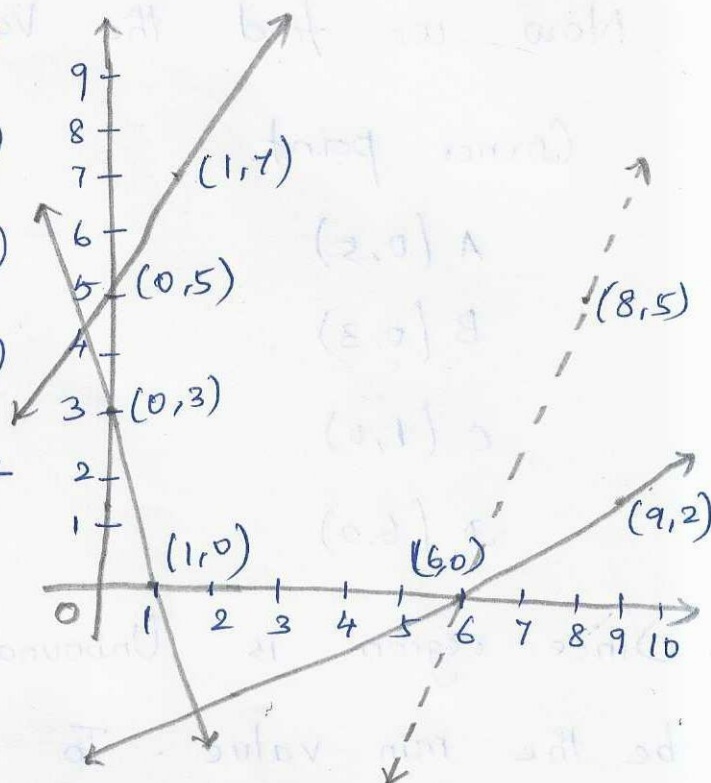
$$2x - y \geq -5 \quad \text{--- (2)}$$

$$3x + y \geq 3 \quad \text{--- (3)}$$

$$2x - 3y \leq 12 \quad \text{--- (4)}$$

$$x \geq 0, y \geq 0 \quad \text{--- (5)}$$

First, Let us find feasible region of the system of inequalities (2), (3), (4), (5)



from (2) $2x - y \geq -5$ $2x - y = -5$

$x = 0$ $x = 1$ $(0, 0)$

$y = 5$ $y = 7$ $0 \geq -5$

$(0, 5)$ $(1, 7)$

from (3) $3x + y \geq 3$

$3x + y = 3$

$x = 0$ $x = 1$

$y = 3$ $y = 0$

$(0, 3)$ $(1, 0)$

$3x + y \geq 3$

$0 + 0 \geq 3$

$0 \geq 3$

from (4) $2x - 3y \leq 12$

$x = 6$ $x = 9$

$y = 0$ $y = 2$

$2x - 3y = 12$

$(0, 0)$

$0 \leq 12$

$$(6,0) \quad (9,2)$$

We observe that feasible region is Unbounded.

Now we find the value of Z at corner points.

Corner point

$$Z = -50x + 20y$$

A (0,5)

$$100$$

B (0,3)

$$60$$

C (1,0)

$$-50$$

D (6,0)

$$-300$$

← Minimum

Since region is Unbounded, -300 may or may not be the min value. To decide this issue, we graph,

$$-50x + 20y < -300$$

$$-5x + 2y < -30$$

$$x = 6$$

$$x = 8$$

$$y = 0$$

$$y = 5$$

$$-5x + 2y < -30$$

$$0 < -30$$

Now, we can see that the resulting open half and feasible region has point in common.

So, -300 will not be the min. value

Therefore there is no Min value of Z .

Ex: 12.1

Solve the following Linear programming problems graphically:

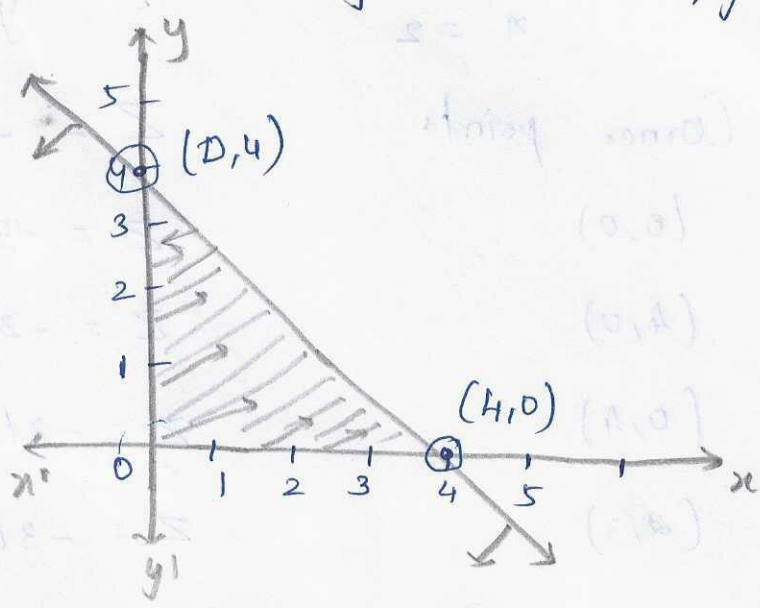
1. Maximise $Z = 3x + 4y$

Subject to the Constraints: $x + y \leq 4$, $x \geq 0$, $y \geq 0$

$x + y \leq 4$

$x \quad 0 \quad 4$

$y \quad 4 \quad 0$



Corner points

$(0, 0)$

$Z = 3x + 4y$

$Z = 0 + 0 = 0$

$(4, 0)$

$Z = 12 + 0 = 12$

$(0, 4)$

$Z = 0 + 16 = 16$

Maximum Value is 16 at $(0, 4)$

2. Minimise $Z = -3x + 4y$

Subject to $x + 2y \leq 8$, $3x + 2y \leq 12$, $x \geq 0$, $y \geq 0$

$x + 2y \leq 8$

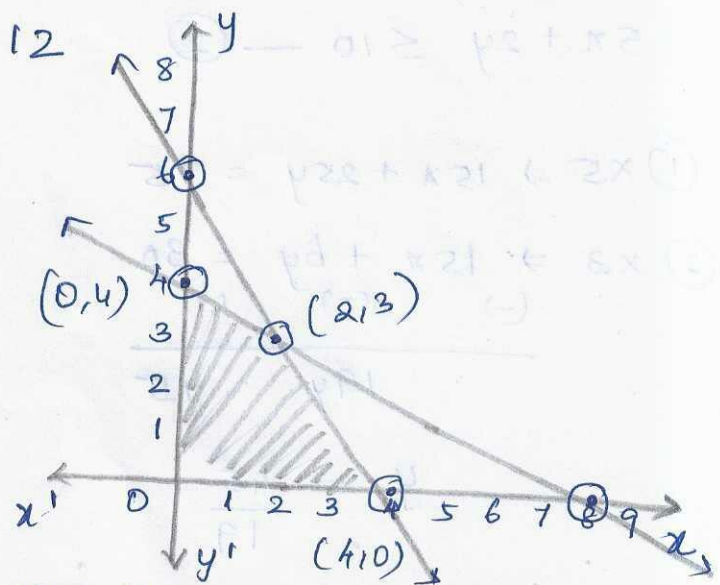
$3x + 2y \leq 12$

$x \quad 0 \quad 8$

$x \quad 0 \quad 4$

$y \quad 4 \quad 0$

$y \quad 6 \quad 0$



$$3x + 2y = 12$$

$$\begin{array}{r} x + 2y = 8 \\ (-) \quad (-) \quad (-) \end{array}$$

$$2x = 4$$

$$x = 2$$

$$2 + 2y = 8$$

$$2y = 8 - 2$$

$$2y = 6$$

$$y = 3$$

Corner points

$$(0,0)$$

$$(4,0)$$

$$(0,4)$$

$$(2,3)$$

$$Z = -3x + 4y$$

$$Z = -3(0) + 4(0) = 0$$

$$Z = -3(4) + 4(0) = -12$$

$$Z = -3(0) + 4(4) = 16$$

$$Z = -3(2) + 4(3) = 6$$

Minimum Value is -12 at $(4,0)$

3. Maximise $Z = 5x + 3y$

Subject to $3x + 5y \leq 15$, $5x + 2y \leq 10$, $x \geq 0$, $y \geq 0$

$$3x + 5y \leq 15$$

$$5x + 2y \leq 10$$

$$x \quad 0 \quad 5$$

$$x \quad 0 \quad 2$$

$$y \quad 3 \quad 0$$

$$y \quad 5 \quad 0$$

$$3x + 5y \leq 15 \text{ --- (1)}$$

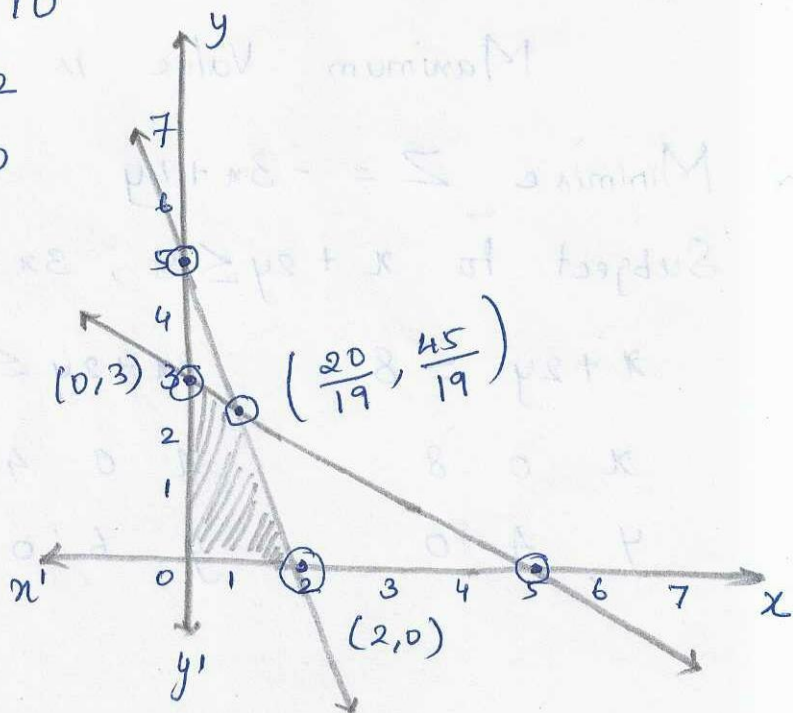
$$5x + 2y \leq 10 \text{ --- (2)}$$

$$\textcircled{1} \times 5 \Rightarrow 15x + 25y = 75$$

$$\textcircled{2} \times 3 \Rightarrow 15x + 6y = 30$$

$$\begin{array}{r} (-) \quad (-) \quad (-) \\ \hline 19y = 45 \end{array}$$

$$y = \frac{45}{19}$$



$$(1) \times 2 \Rightarrow 6x + 10y = 30$$

$$(2) \times 5 \Rightarrow \begin{array}{r} 25x + 10y = 50 \\ (-) \quad (-) \quad (-) \end{array}$$

$$-19x = -20$$

$$x = 20/19$$

Corner points

$$(0,0)$$

$$Z = 5x + 3y$$

$$Z = 5(0) + 3(0) = 0$$

$$(2,0)$$

$$Z = 5(2) + 3(0) = 10$$

$$(0,3)$$

$$Z = 5(0) + 3(3) = 9$$

$$\left(\frac{20}{19}, \frac{45}{19}\right)$$

$$Z = 5\left(\frac{20}{19}\right) + 3\left(\frac{45}{19}\right)$$

$$= \frac{100}{19} + \frac{135}{19}$$

$$= \frac{235}{19}$$

$$Z_{\max} = \frac{235}{19} \text{ at } \left(\frac{20}{19}, \frac{45}{19}\right)$$

4. Minimise $Z = 3x + 5y$

Such that $x + 3y \geq 3$, $x + y \geq 2$, $x, y \geq 0$

$$x + 3y \geq 3$$

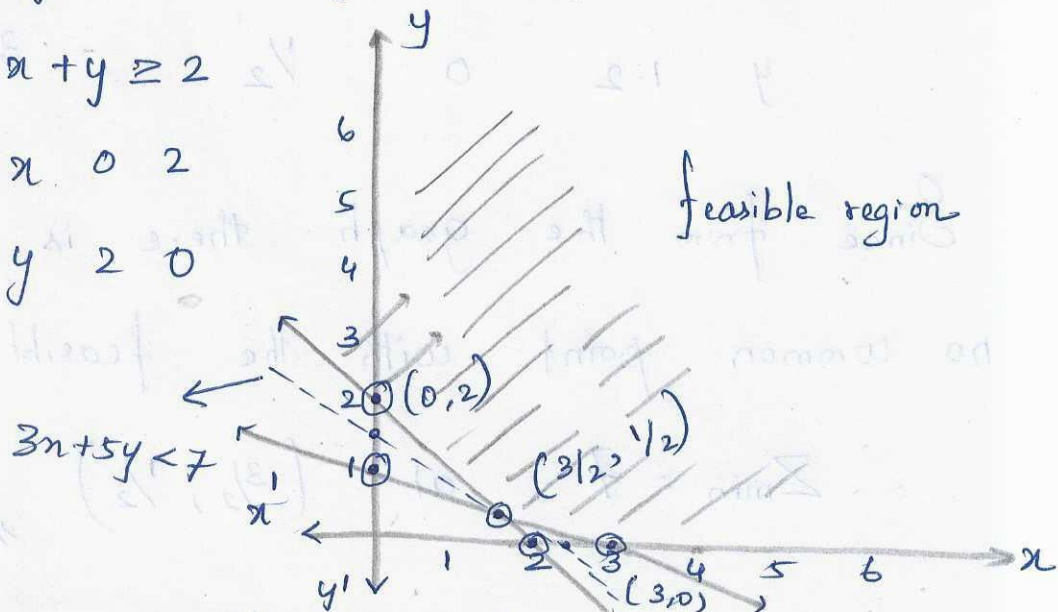
$$x + y \geq 2$$

$$x \quad 0 \quad 3$$

$$x \quad 0 \quad 2$$

$$y \quad 1 \quad 0$$

$$y \quad 2 \quad 0$$



$$\begin{array}{r} x + 3y = 3 \\ x + y = 2 \\ \hline (-) \quad (-) \quad (-) \end{array}$$

$$2y = 1$$

$$y = 1/2$$

$$x + 1/2 = 2$$

$$x = 2 - 1/2$$

$$x = \frac{4-1}{2}$$

$$x = 3/2$$

Corner points

$$(0, 2)$$

$$(3, 0)$$

$$(3/2, 1/2)$$

$$Z = 3x + 5y$$

$$Z = 3(0) + 5(2) = 10$$

$$Z = 3(3) + 5(0) = 9$$

$$Z = 3(3/2) + 5(1/2)$$

$$= 9/2 + 5/2$$

$$= 14/2 = 7 \text{ (Minimum)}$$

As the region is unbounded.

So, 7 may or may not be the minimum

$$3x + 5y < 7$$

$$5y = 7$$

$$x = 0 \quad y = 7/5 = 1.2$$

$$y = 1.2 \quad x = 0 \quad 1/2$$

$$3x = 7$$

$$x = 7/3 = 2.3$$

Since from the graph there is

no common point with the feasible region.

$$\therefore Z_{\min} = 7 \text{ at } (3/2, 1/2)$$

(5)

5. Minimise $Z = 3x + 2y$ Subject to $x + 2y \leq 10$, $3x + y \leq 15$, $x, y \geq 0$

$$x + 2y \leq 10$$

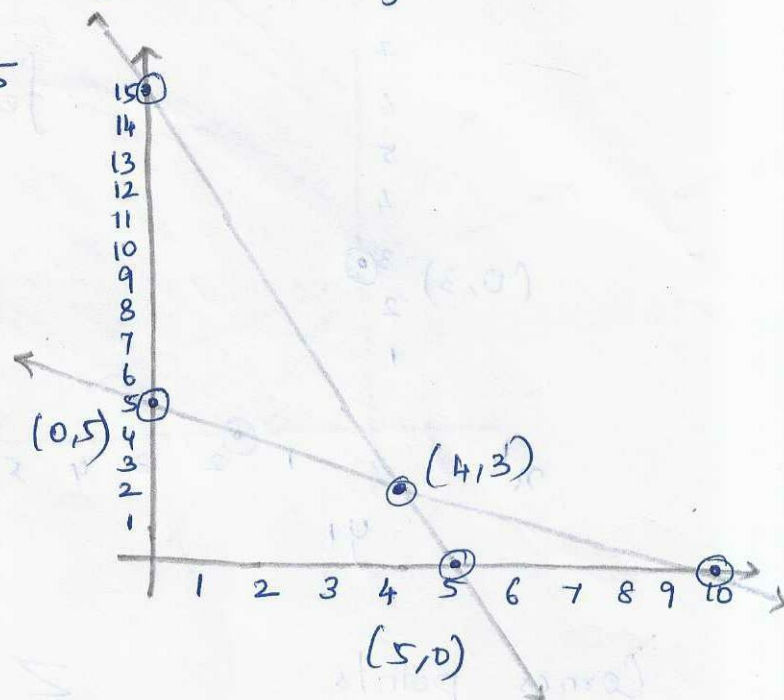
$$3x + y \leq 15$$

$$x \quad 0 \quad 10$$

$$x \quad 0 \quad 5$$

$$y \quad 5 \quad 0$$

$$y \quad 15 \quad 0$$



$$\textcircled{1} \times 3 \Rightarrow 3x + 6y = 30$$

$$\textcircled{2} \Rightarrow \begin{array}{r} 3x + y = 15 \\ (-) \quad (-) \quad (-) \\ \hline 5y = 15 \end{array}$$

$$5y = 15$$

$$y = 3$$

$$x + 2(3) = 10$$

$$x + 6 = 10$$

$$x = 4$$

Corner points

$$(0,0)$$

$$(5,0)$$

$$(0,5)$$

$$(4,3)$$

$$Z = 3x + 2y$$

$$Z = 3(0) + 2(0) = 0$$

$$Z = 3(5) + 2(0) = 15$$

$$Z = 3(0) + 2(5) = 10$$

$$Z = 3(4) + 2(3) = 18$$

$$Z_{\max} = 18 \text{ at } (4,3)$$

6. Minimise $Z = x + 2y$ Subject to $2x + y \geq 3$, $x + 2y \geq 6$, $x, y \geq 0$

$$2x + y \geq 3$$

$$x + 2y \geq 6$$

$$x \quad 0 \quad 1.5$$

$$x \quad 0 \quad 6$$

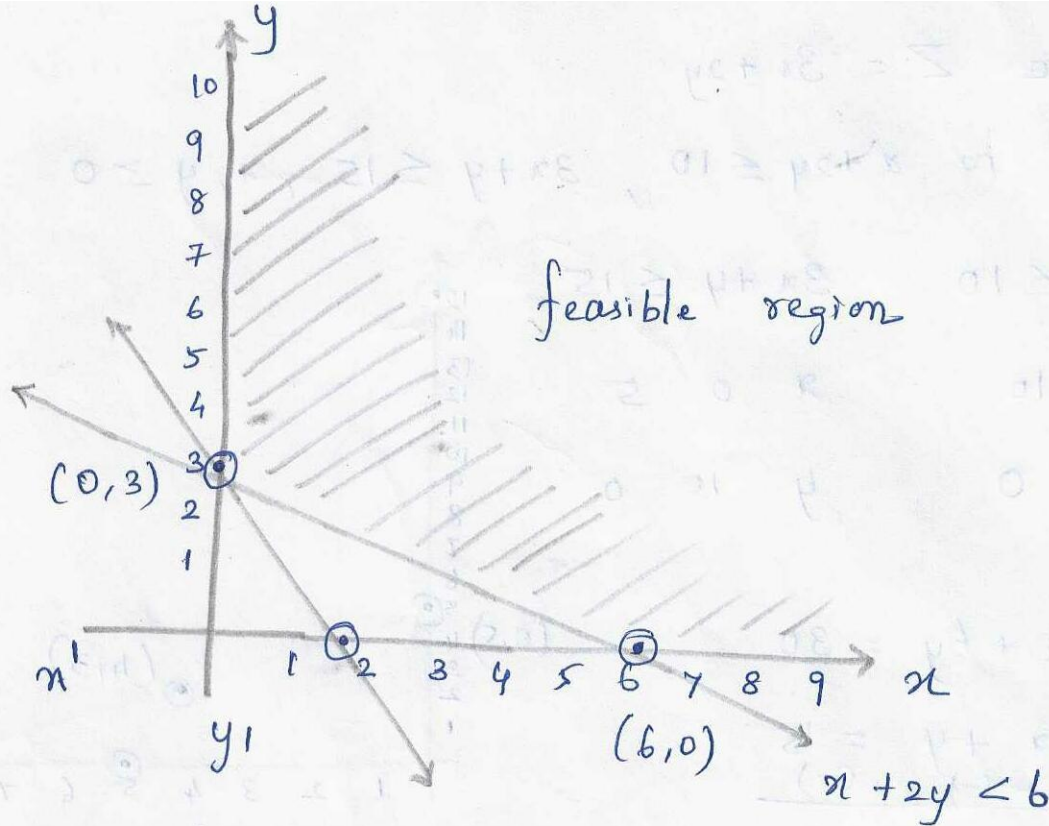
$$y \quad 3 \quad 0$$

$$y \quad 3 \quad 0$$

$$2x + y \geq 3$$

$$2(4) + 4 \geq 3$$

$$12 \geq 3$$



Cornex points

$$(0, 3)$$

$$(6, 0)$$

$$Z = x + 2y$$

$$Z = 0 + 2(3) = 6$$

$$Z = 6 + 2(0) = 6$$

Since the region is unbounded, So we need to
Check 6 is minimum value or not

$$x + 2y <= 6$$

$$x \quad 0 \quad 6$$

$$y \quad 3 \quad 0$$

$$Z_{\min} = 6$$

There is no common region with the graph ^{feasible}
at all points in the line segment joining $(0, 3)$
and $(6, 0)$.

7. Minimise and Maximise $Z = 5x + 10y$

Subject to $x + 2y \leq 120$, $x + y \geq 60$, $x - 2y \geq 0$, $x, y \geq 0$

$$x + 2y \leq 120$$

$$x + y \geq 60$$

$$x - 2y \geq 0$$

$$x \quad 0 \quad 120$$

$$x \quad 0 \quad 60$$

$$x \quad 0 \quad 40$$

$$y \quad 60 \quad 0$$

$$y \quad 60 \quad 0$$

$$y \quad 0 \quad 20$$

L_3 and L_2

$$x + y = 60$$

$$x - 2y = 0$$

$$\begin{array}{r} - \quad (+) \quad - \\ \hline \end{array}$$

$$3y = 60$$

$$y = 60/3$$

$$y = 20$$

$$x + 20 = 60$$

$$x = 60 - 20 = 40$$

$$x = 40$$

L_1 and L_3

$$x + 2y = 120$$

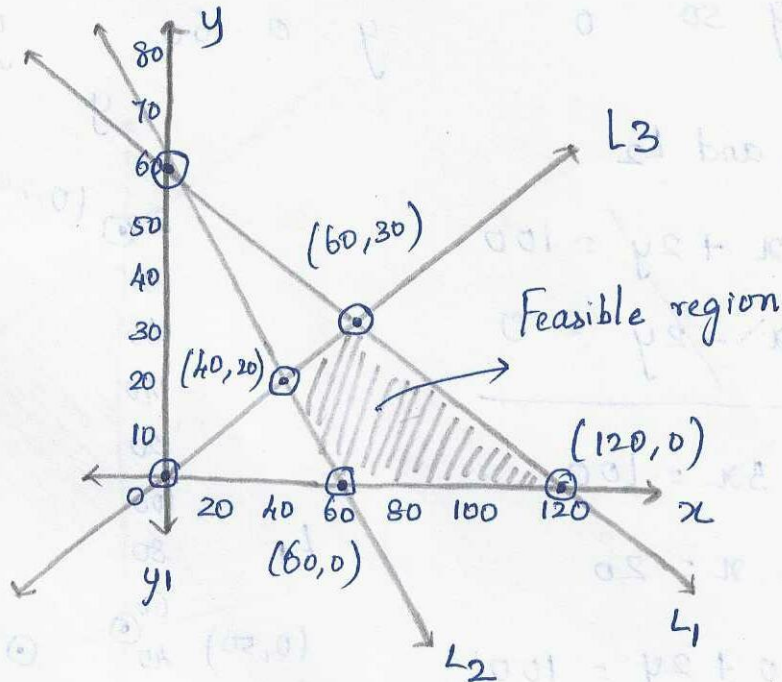
$$x - 2y = 0$$

$$2x = 120$$

$$x = 60$$

$$60 - 2\left(\frac{y}{2}\right) = 0$$

$$x = 120$$



Corner points

$$(60, 0)$$

$$(120, 0)$$

$$(40, 20)$$

$$(60, 30)$$

$$Z = 5x + 10y$$

$$Z = 5(60) + 10(0) = 300$$

$$Z = 5(120) + 10(0) = 600$$

$$Z = 5(40) + 10(20) = 400$$

$$Z = 5(60) + 10(30) = 600$$

$$Z_{\min} = 300 \text{ at } (60, 0)$$

$$Z_{\max} = 600 \text{ at all points joining the line segment } (120, 0) \text{ and } (60, 30)$$

8. Minimise and Maximize $Z = x + 2y$
 Subject to $x + 2y \geq 1000$, $2x - y \leq 0$, $2x + y \leq 200$;
 $x, y \geq 0$.

$$x + 2y \geq 100$$

$$x \quad 0 \quad 100$$

$$y \quad 50 \quad 0$$

$$2x - y \leq 0$$

$$x \quad 0 \quad 30$$

$$y \quad 0 \quad 60$$

$$2x + y \leq 200$$

$$x \quad 0 \quad 100$$

$$y \quad 200 \quad 0$$

L_1 and L_2

$$x + 2y = 100$$

$$4x - 2y = 0$$

$$5x = 100$$

$$x = 20$$

$$20 + 2y = 100$$

$$2y = 100 - 20$$

$$2y = 80$$

$$y = 40$$

L_2 and L_3

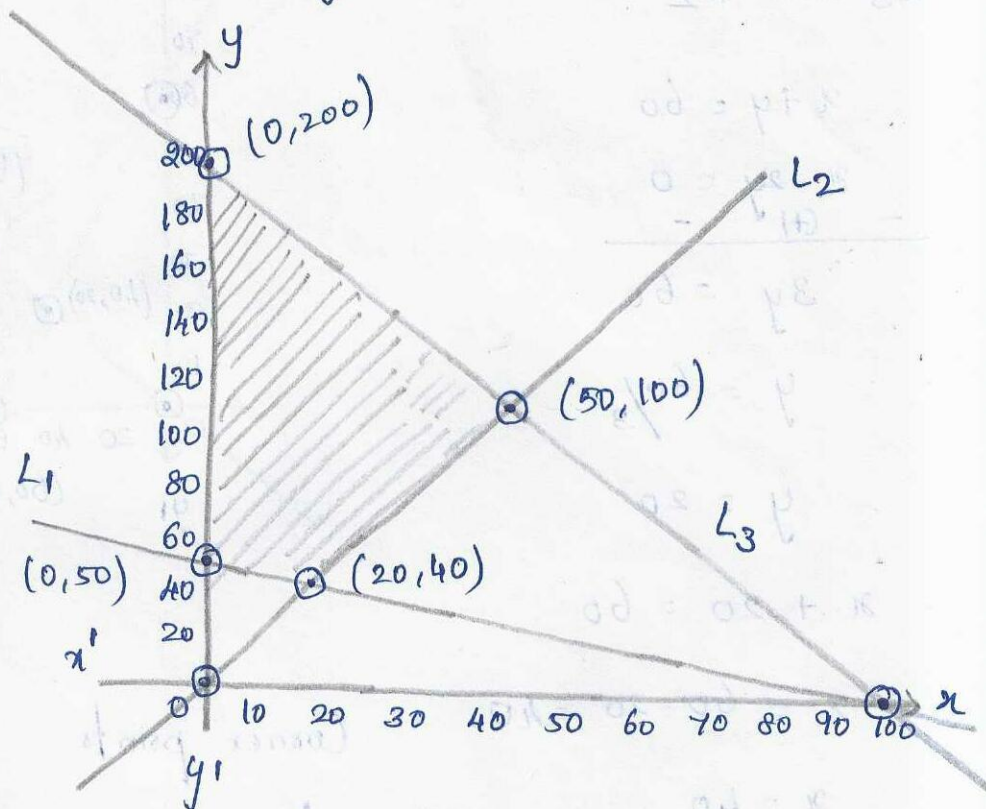
$$2x - y = 0$$

$$2x + y = 200$$

$$4x = 200$$

$$x = 50$$

$$2(50) - y = 0 \quad \therefore y = 100$$



Corner points

$$(0, 200)$$

$$(0, 50)$$

$$(20, 40)$$

$$(50, 100)$$

$$Z = x + 2y$$

$$Z = 0 + 2(200) = 400$$

$$Z = 0 + 2(50) = 100$$

$$Z = 20 + 2(40) = 100$$

$$Z = 50 + 2(100) = 250$$

$$Z_{\max} = 400 \text{ at } (0, 200)$$

$$Z_{\min} = 100 \text{ at all points joining } (0, 50) \text{ and } (20, 40)$$

9. Maximise $Z = -x + 2y$, Subject to the constraints;

$$x \geq 3, x + y \geq 5, x + 2y \geq 6, y \geq 0$$

$$x \geq 3 \quad x + y \geq 5 \quad x + 2y \geq 6$$

$$\begin{array}{ccc} x & 0 & 5 \\ y & 5 & 0 \end{array}$$

$$\begin{array}{ccc} x & 0 & 6 \\ y & 3 & 0 \end{array}$$

L_2 and L_3

L_1

$$x + y = 5$$

$$x = 3$$

$$\begin{array}{r} x + 2y = 6 \\ (-) \quad (-) \quad (-) \\ \hline \end{array}$$

$$x + y = 5$$

$$-y = -1$$

$$3 + y = 5$$

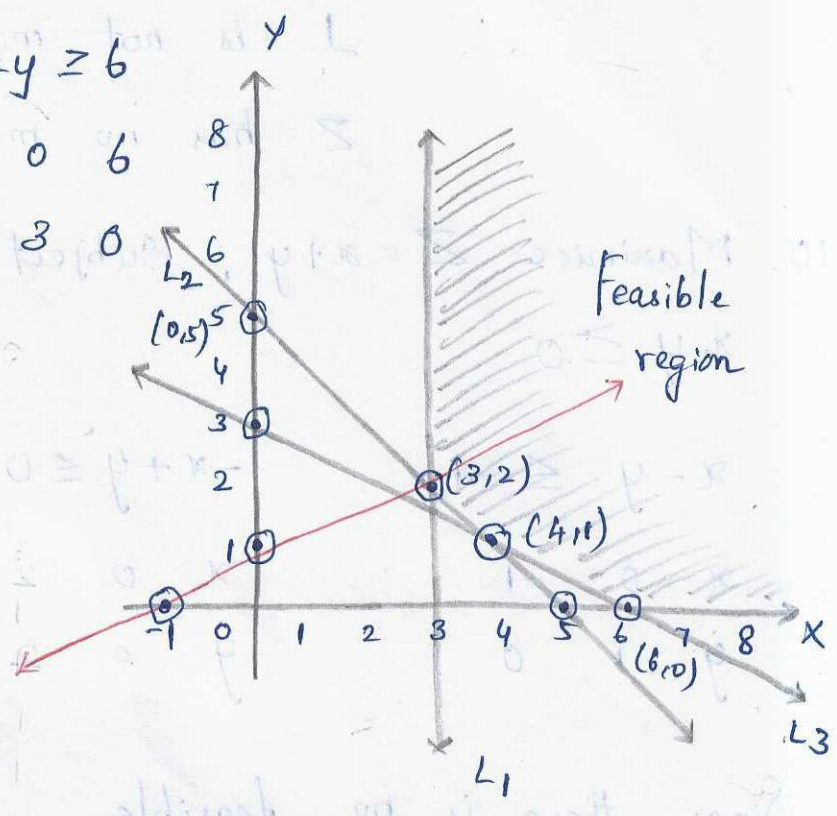
$$y = 1$$

$$y = 5 - 3$$

$$x + 1 = 5$$

$$y = 2$$

$$x = 4$$



Corner points

$$(3, 2)$$

$$Z = -x + 2y$$

$$(4, 1)$$

$$Z = -3 + 2(2) = -3 + 4 = 1$$

$$(6, 0)$$

$$Z = -4 + 2(1) = -4 + 2 = -2$$

$$Z = -6 + 2(0) = -6$$

Since the region is unbounded, so I may or may not be maximum value.

$$-x + 2y > 1$$

$$2y = 1$$

$$\begin{array}{ccc} x & 0 & -1 \end{array}$$

$$y = 0.5$$

$$\begin{array}{ccc} y & 0.5 & 0 \end{array}$$

$$-x = 1$$

$$x = -1$$

Graph of $-x + 2y \geq 1$ has common points with the feasible region.

$\therefore 1$ is not max Value
 Z has no max Value.

10. Maximise $Z = x + y$, Subject to $x - y \leq -1$, $-x + y \leq 0$,
 $x, y \geq 0$

$$x - y \leq -1 \quad -x + y \leq 0$$

x	0	-1	x	0	2
y	1	0	y	0	2

Since there is no feasible region, No solution.

